

501

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 #define mp make_pair
4 #define pb push_back
5 typedef long long ll;
6
7 priority_queue<int> p;
8 priority_queue<int,vector<int>,greater<int> > q;
9
10 int a[30005],b[30005];
11
12 int main(){
13     int T,t;cin>>T;
14     while(T--){
15         int n,m;
16         cin>>n>>m;
17         while(!q.empty()) q.pop();
18         while(!p.empty()) p.pop();
19
20         for(int i=1;i<=n;i++) scanf("%d",&a[i]);
21         for(int i=1;i<=m;i++) scanf("%d",&t),b[t]++;
22
23         int c=0;
24         for(int i=1;i<=n;i++){
25             p.push(a[i]);
26             q.push(p.top());
27             p.pop();
28
29             while(b[i]){
30                 int u=q.top(); q.pop();
31                 p.push(u);
32                 printf("%d\n",u);
33                 b[i]--;
34             }
35         }
36
37         if(T) puts("");
38     }
39 }

```

Messages

[17:47:59] [Compile Warnings]
 /tmp/CP_Editor-vuJNQ/sol.cpp: In function 'int main()':
 /tmp/CP_Editor-vuJNQ/sol.cpp:23:13: warning: unused variable 'c' [-Wunused-variable]
 23 | int c=0;
 |

[17:47:59] [Runner[1]] [Execution has started]
 [17:47:59] [Runner[1]] [Execution for test case #1 has finished in 0ms]

Test Cases: 0 / 1

Checker: Ignore trailing spaces

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	1		3		3	
	7 4		3		3	
	3 1 -4 2 8 -1000 2		1		1	
	1 2 6		2		2	

793

```

7     return par[r] = r ? r : par[r] = find(par[r]);
8 }
9
10 int main() {
11     //freopen("in", "r", stdin);
12
13     int t, n, m, u, v;
14     char ch;
15
16     for (scanf("%d", &t); t--;) {
17         scanf("%d", &n);
18
19         for (int i = 1; i <= n; i++)
20             par[i] = i;
21
22         getchar();
23
24         int a(0), b(0);
25
26         while (ch = getchar()) {
27             if (!(ch == 'c' || ch == 'q'))
28                 break;
29
30             scanf("%d %d", &u, &v);
31
32             u = find(u);
33             v = find(v);
34
35             if (ch == 'c' && u != v)
36                 par[u] = v;
37             else if (ch == 'q')
38                 (u == v) ? a++ : b++;
39
40             getchar();
41         }
42
43         printf("%d,%d\n", a, b);
44
45         if (t != 0)
46             puts("");
47     }
48 }

```

Messages

26 | while (ch = getchar()) {

 /tmp/CP_Editor-vuJNQ/sol.cpp:13:15: warning: unused variable 'n' [-Wunused-variable]
 13 | int t, n, m, u, v;
 |

[17:57:05] [Runner[1]] [Execution has started]
 [17:57:05] [Runner[1]] [Execution for test case #1 has finished in 2ms]

Test Cases: 0 / 1

Checker: Ignore trailing spaces

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	2		1,2		1,2	
	10		2,0		2,0	
	c 1 5					
	c 2 7					
	q 7 1					
	c 3 9					
	q 9 6					
	c 2 5					
	q 7 5					
	1					
	q 1 1					
	c 1 1					
	q 1 1					

798

The screenshot shows a C++ IDE with a DFS solution for problem 798. The code is as follows:

```

32     return 1;
33 }
34
35 int dfs(int t, int num){
36     if(num==tile_num) return 1;
37
38     int x=t/r, y=t%r, ans=0;
39     if(board[x][y]) return dfs(t+1, num);
40
41     for(int i=0; i<n; i++){
42         if(!N[i]) continue;
43
44         if(CanPut(t, i, 0)){
45             Put(t, i, 0); N[i]--;
46             ans+=dfs(t+1, num+1);
47             Put(t, i, 0); N[i]++;
48         }
49
50         if(CanPut(t, i, 1)){
51             Put(t, i, 1); N[i]--;
52             ans+=dfs(t+1, num+1);
53             Put(t, i, 1); N[i]++;
54         }
55     }
56     return ans;
57 }
58
59 int main(){
60     while(cin>>r>>c>>n){
61         init();
62
63         for(int i=0; i<n; i++){
64             cin>>k>>x>>y;
65             X.pb(x), Y.pb(y), N.pb(k);
66             tile_num+=k;
67         }
68
69         cout<<dfs(0, 0)<<'\n';
70     }
71 }

```

The Messages panel shows the following output:

```

[18:36:20] [Compiler] [Compilation has started]
[18:36:22] [Compiler] [Compilation has finished]
[18:36:22] [Runner[1]] [Execution has started]
[18:36:22] [Runner[1]] [Execution for test case #1 has finished in 0ms]

```

The Test Cases panel shows the following results:

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	3 2 2		11		11	
	2 1 1		56		56	
	2 1 2					
	5 2 2					
	2 1 1					
	4 1 2					

908

The screenshot shows a C++ IDE with a solution for problem 908. The code is as follows:

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 vector<int>F;
4 #define f first
5 #define s second
6 int find(int a){
7     return F[a]=F[a]==a?a:find(F[a]);
8 }
9 int main(){
10     int t,n,a,b,w,k=0;
11     while(cin>>t){
12         int olds<0, news=0;
13         F=vector<int>(t+1);
14         for(int i=0; i<=t; i++)F[i]=i;
15         for(int i=1; i<t; i++)cin>>a>>b>>w, olds+=w;
16         vector<pair<int, pair<int, int>>>ans;
17         cin>>n;
18         for(int i=0; i<n; i++){
19             cin>>a>>b>>w;
20             ans.push_back({w, {a, b}});
21         }
22         cin>>n;
23         for(int i=0; i<n; i++){
24             cin>>a>>b>>w;
25             ans.push_back({w, {a, b}});
26         }
27         sort(ans.begin(), ans.end());
28         for(auto&i:ans){
29             int f1=find(i.s.f), f2=find(i.s.s);
30             if(f1!=f2){
31                 news+=1.f;
32                 F[f1]=F[f2];
33             }
34         }
35         cout<<(k++?"\n":"")<<olds<<'\n'<<news<<'\n';
36     }
37 }

```

The Messages panel shows the following output:

```

[18:45:11] [Compiler] [Compilation has started]
[18:45:13] [Compiler] [Compilation has finished]
[18:45:13] [Runner[1]] [Execution has started]
[18:45:13] [Runner[1]] [Execution for test case #1 has finished in 0ms]

```

The Test Cases panel shows the following results:

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	5		20		20	
	1 2 5		17		17	
	1 3 5					
	1 4 5					
	1 5 5					
	1					
	2 3 2					
	6					
	1 2 5					
	1 3 5					
	1 4 5					
	1 5 5					
	3 4 8					
	4 5 8					

926

```

1 #include<cstdio>
2 #include<cstring>
3 #define ll long long
4 int p,sx,sy,ex,ey,n,t,a[35][35][3];ll f[35][35];
5 int main(){
6     scanf("%d",&t);
7     while (t--){
8         memset(f,0,sizeof(f));memset(a,0,sizeof(a));
9         scanf("%d%d%d%d%d%d",&p,&sx,&sy,&ex,&ey,&n);int c,x,y;
10        for (int i=1;i<=n;i++){
11            scanf("%d%d",&bx,&by);c=getchar();c=getchar();
12            if (c=='N') a[x][y][1]=i;
13            else if (c=='E') a[x][y][2]=1;
14            else if (c=='S') a[x][y-1][1]=i;
15            else a[x-1][y][2]=i;
16        }
17        f[sx][sy]=1;
18        for (int i=sx;i<=ex;i++){
19            for (int j=sy;j<=ey;j++){
20                if (sx==i&&sy==j)continue;
21                if (!a[i][j][1])f[i][j]+=f[i-1][j];
22                if (!a[i][j][2])f[i][j]+=f[i][j-1];
23            }
24            printf("%lld\n",f[ex][ey]);
25        }
26        return 0;
27 }
28

```

Messages

[18:48:02] [Compiler] [Compilation has started]
[18:48:02] [Compiler] [Compilation has finished]
[18:48:02] [Runner[1]] [Execution has started]
[18:48:02] [Runner[1]] [Execution for test case #1 has finished in 0ms]

Test Cases 0 / 1

Checker: Ignore trailing spaces

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	2		3		3	
	3		6		6	
	1 1					
	3 3					
	2					
	2 3 5					
	2 2 W					
	3					
	1 1					
	3 3					
	0					

1632

```

1 #include<bits/stdc++.h>
2 using namespace std;
3 int main(){
4     int n;
5     while(cin>>n){
6         int dp[n][n][2]={}, p[n], t[n], sum;
7         for(int i=0;i<n;i++)cin>>p[i]>>t[i];
8         for(int i=n-2;i>=0;i--)for(int j=i+1;j<n;j++){
9             dp[i][j][0]=min(dp[i+1][j][0]+p[i+1]-p[i],dp[i+1][j][1]+p[j]-p[i]);
10            dp[i][j][1]=min(dp[i][j-1][0]+p[j]-p[i],dp[i][j-1][1]+p[j]-p[j-1]);
11            if(dp[i][j][0]>=t[i])dp[i][j][0]=999999;
12            if(dp[i][j][1]>=t[j])dp[i][j][1]=999999;
13        }
14        sum=min(dp[0][n-1][0],dp[0][n-1][1]);
15        if(sum==999999)cout<<"No solution\n";
16        else cout<<sum<<endl;
17    }
18 }
19

```

Messages

[18:49:47] [Compiler] [Compilation has started]
[18:49:49] [Compiler] [Compilation has finished]
[18:49:49] [Runner[1]] [Execution has started]
[18:49:49] [Runner[1]] [Execution for test case #1 has finished in 0ms]

Test Cases 0 / 1

Checker: Ignore trailing spaces

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	5		11		11	
	1 3		No solution		No solution	
	3 1					
	5 8					
	8 19					
	10 15					
	5					
	1 5					
	2 1					
	3 4					
	4 2					
	5 3					

1757

```

8 bool query(string s, string t){
9
10     if(s.size() != t.size()) return false;
11     int sz = s.size();
12     for(int i = 0; i < sz; i++){
13         if(!can[s[i] - 'a'][t[i] - 'a']) return false;
14     }
15
16     return true;
17 }
18
19 void solve(int m, int n){
20     memset(can, 0, sizeof can);
21     for(int i = 0; i < 26; i++) can[i][i] = 1;
22
23     for(int i = 0; i < m; i++){
24         char c, d;
25         cin >> c >> d;
26         can[c - 'a'][d - 'a'] = 1;
27     }
28     for(int k = 0; k < 26; k++){
29         for(int i = 0; i < 26; i++){
30             for(int j = 0; j < 26; j++){
31                 can[i][j] |= can[i][k] & can[k][j];
32             }
33         }
34     }
35     string s, t;
36     for(int i = 0; i < n; i++){
37         cin >> s >> t;
38         query(s, t) ? cout << "yes\n" : cout << "no\n";
39     }
40 }
41
42 int main()
43 {
44     int m, n;
45     while(cin >> m >> n){
46         solve(m, n);
47     }
48 }

```

Messages

[18:51:28] [Compiler] [Compilation has started]
 [18:51:29] [Compiler] [Compilation has finished]
 [18:51:30] [Runner[1]] [Execution has started]
 [18:51:30] [Runner[1]] [Execution for test case #1 has finished in 0ms]

Test Cases 0 / 1

Checker: Ignore trailing spaces

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	9 5		yes		yes	
	c t		no		no	
	i r		no		no	
	k p		yes		yes	
	o c		yes		yes	
	r o		yes		yes	
	t e		no		no	
	t f		yes		yes	
	u h					
	w p					
	we we					
	can the					
	work people					
	it of					
	out the					
	3 3					
	a c					
	b a					
	a b					
	aaa abc					
	abc aaa					
	acm bcm					

10000

```

1 #include <bits/stdc++.h>
2 using namespace std;
3 int dp[120], n, ans[120];
4 vector<int> vc[120];
5
6 int dp_func(int v)
7 {
8     if (dp[v] != -1) return dp[v];
9     int sum = 0, i;
10    ans[v] = v;
11    for (i = 0; i < vc[v].size(); i++) {
12        int w = vc[v][i];
13        if (sum == dp_func(w) + 1) {
14            ans[v] = min(ans[v], ans[w]);
15        }
16        if (sum < dp_func(w) + 1) {
17            sum = dp_func(w) + 1;
18            ans[v] = ans[w];
19        }
20    }
21    return dp[v] = sum;
22 }
23
24 int main()
25 {
26     int n, m, i, j, k, l, casio = 1;
27     while (scanf("%d", &n) & n != 0)
28     {
29         scanf("%d", &m);
30         while (scanf("%d%d", &k, &l)) {
31             if (k == 0 && l == 0) break;
32             vc[k].push_back(l);
33         }
34         memset(dp, -1, sizeof dp);
35         l = dp_func(m);
36         printf("Case %d: The longest path from %d has length %d, finishing at %d.\n", casio, m, l, ans[m]);
37         for (i = 1; i <= n; i++) vc[i].clear();
38     }
39 }

```

Messages

27 | while (scanf("%d", &n) & n != 0)
 /tmp/CP-Editor-m05e/i/sol.cpp:26:18: warning: unused variable 'j' [-Wunused-variable]
 26 | int n, m, i, j, k, l, casio = 1;
 | ^

[18:54:11] [Runner[1]] [Execution has started]
 [18:54:11] [Runner[1]] [Execution for test case #1 has finished in 3ms]

Test Cases 0 / 1

Checker: Ignore trailing spaces

✓	Input #1	Run	Output #1	AC	Expected #1	Del
	2		Case 1: The longest path from 2 has length 2, finishing at 2.		Case 1: The longest path from 2 has length 2, finishing at 2.	
	1					
	1 2		Case 2: The longest path from 1 has length 2, finishing at 2.		Case 2: The longest path from 1 has length 2, finishing at 2.	
	0 0					
	5		Case 3: The longest path from 5 has length 5, finishing at 5.		Case 3: The longest path from 5 has length 5, finishing at 5.	
	3					
	1 2					
	3 5					
	3 1					
	2 4					
	4 5					
	0 0					
	5					
	5					
	5 1					
	5 2					
	5 3					
	5 4					
	4 1					
	4 2					
	0 0					
	0					

10033

```
18 while (_o--){
19     for (int i = 0; i < 1000; i++) R[i] = 0;
20
21     int n = 0;
22     while (getline(cin, s) && !s.empty()) {
23         if (s.length() ≥ 3) {
24             R[n] = (s[0] - '0') * 100 + (s[1] - '0') * 10 + (s[2] - '0');
25         } else if (s.length() == 1) {
26             R[n] = (s[0] - '0') * 100;
27         }
28         n++;
29     }
30
31     for (int i = 0; i < 10; i++) A[i] = 0;
32
33     int ans = 0;
34     int i = 0;
35     while (true) {
36         ans++;
37         if (R[i] == 100) break;
38
39         int typ = R[i] / 100;
40         int a = (R[i] % 100) / 10;
41         int b = R[i] % 10;
42
43         if (typ == 2) A[a] = b;
44         else if (typ == 3) A[a] = (A[a] + b) % 1000;
45         else if (typ == 4) A[a] = (A[a] * b) % 1000;
46         else if (typ == 5) A[a] = A[b];
47         else if (typ == 6) A[a] = (A[a] + A[b]) % 1000;
48         else if (typ == 7) A[a] = (A[a] * A[b]) % 1000;
49         else if (typ == 8) A[a] = R[A[b]];
50         else if (typ == 9) R[A[b]] = A[a];
51         else if (typ == 0) {
52             if (A[b] ≠ 0) {
53                 i = A[a];
54                 continue;
55             }
56         }
57         i++;
58     }
59 }
```

Messages

Clear C++

[19:03:34] [Compiler] [Compilation has started]
[19:03:35] [Compiler] [Compilation has finished]
[19:03:35] [Runner[1]] [Execution has started]
[19:03:35] [Runner[1]] [Execution for test case #1 has finished in 3ms]

Test Cases 0 / 1 Add Test More

Checker: Ignore trailing spaces Add Checker

✓	Input #1	Run	Output #1	AC	Expected #1	Del
1			16		16	
299						
492						
495						
399						
492						
495						
399						
283						
279						
689						
078						
100						
000						
000						
000						

line 43, Column 41 Compile Run Compile and Run

10042

```
1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5 int t, n;
6
7 int sum (int x) {
8     int res = 0;
9     while (x) res += (x % 10), x /= 10;
10    return res;
11 }
12
13 bool check (int x) {
14     int cur = sum(x), res = 0;
15     for (int i = 2; i * i ≤ x; ++i) {
16         while (x % i == 0) {
17             res += sum(i), x /= i;
18         }
19     }
20     if (res == 0) return 0;
21     if (x > 1) res += sum(x);
22     return cur == res;
23 }
24
25 int main() {
26     scanf("%d", &t);
27     while (t--) {
28         scanf("%d", &n);
29         ++n;
30         while (!check(n)) ++n;
31         printf("%d\n", n);
32     }
33     return 0;
34 }
35
```

Messages

Clear C++

[19:16:57] [Compiler] [Compilation has started]
[19:16:59] [Compiler] [Compilation has finished]
[19:16:59] [Runner[1]] [Execution has started]
[19:16:59] [Runner[1]] [Execution for test case #1 has finished in 0ms]

Test Cases 0 / 1 Add Test More

Checker: Ignore trailing spaces Add Checker

✓	Input #1	Run	Output #1	AC	Expected #1	Del
1	4937774		4937775		4937775	

line 35, Column 1 Compile Run Compile and Run